

Two True Maps — Filled Grids

Teacher copy. Pair 02 and Pair 03, both sides of each pair. For discussion, not for handing out.

Pair 02 · The Two Bar Charts

EMPLOYMENT RATE, 2020–2025 · IDENTICAL NUMBERS IN BOTH CHARTS · ROUGHLY 92% TO 98%

CHART A — Y-AXIS 90% TO 100%

01 Includes	Six bars, 2020 to 2025, titled Employment Rate. The axis runs 90% to 100% and is honestly labelled. The 2022 bar is a stub; the 2025 bar towers over it. Differences between years are large and obvious.
02 Leaves out	The bottom 90% of the scale. With the baseline cut away, bar height no longer stands for quantity — it stands only for the part above 90%. The 2022 bar is not one fifth of the 2025 bar, but it looks like it.
03 Right for	A labour analyst who already knows employment sits in the nineties and whose job is to spot a half-point move. For that person the space from 0 to 90 is dead paper, and Chart A is the only one of the two charts where the 2022 dip is visible at all.
04 Misleads	A voter deciding whether there was an employment crisis in 2022. Criterion: proportion — the size of the change measured against the size of the whole. Chart A keeps the direction and the rank order of the years, and throws proportion away. The voter's question is purely a proportion question, so the one property they need is the one the chart gave up.

CHART B — Y-AXIS 0% TO 100%

01 Includes	The same six years and the same numbers, roughly 92% to 98%. The axis runs 0% to 100%. Six tall bars, almost level with each other.
02 Leaves out	Nothing about the data — but the differences between years are squeezed almost flat. The 2022 drop is about six points and is nearly invisible at this scale. The change is in the numbers and absent from the picture.
03 Right for	Someone asking how many people are employed overall, who needs to know the level is high rather than low. Criterion: proportion, and Chart B keeps it.
04 Misleads	An analyst looking for the year something went wrong. Criterion: resolution — being able to see small differences between years. Chart B flattens 2022 into the row, so the analyst reads “nothing happened” off a chart that contains something happening.

The point students should reach: **both** charts mislead somebody. Chart B is not the honest one and Chart A is not the lying one. Each keeps a property the other gives up.

Pair 03 · The Two Photographs

THE MARCH ON WASHINGTON FOR JOBS AND FREEDOM · 28 AUGUST 1963 · SAME DAY, SAME EVENT

PHOTOGRAPH A — THE AERIAL

01 Includes	The whole crowd seen from above: two to three hundred thousand people, and how far down the Mall they stretch. You can see the size of the thing.
02 Leaves out	Every face. Every name. Every reason anyone came, and what it cost them to travel. From that height a person is a dot.
03 Right for	A historian estimating turnout, or anyone arguing that the march was a mass movement and not a fringe event. Criterion: magnitude, which no other kind of photograph preserves as well.
04 Misleads	Someone asking who was at the march. Criterion: individual identity . The aerial cannot give a single name, and it makes a quarter of a million people look like one thing with one motive, when they came for a quarter of a million reasons.

PHOTOGRAPH B — THE CLOSE-UP

01 Includes	One twelve-year-old girl, her face filling the frame, and her expression. One person, completely.
02 Leaves out	Everyone else. Nothing in this photograph tells you whether two hundred people came or two hundred thousand. There is no scale in it at all.
03 Right for	Someone asking what it meant to be there — age, feeling, commitment, none of which survives in the aerial. Criterion: individual identity.
04 Misleads	Someone estimating the size of the march. Criterion: magnitude . A single face fits any crowd size, so a knower who reads scale off this photograph can be wrong by a factor of a thousand and never notice.

Worth raising at the end: Edith Lee-Payne did not know the photograph of her existed until 2008, forty-five years later. The photograph was right for everyone except its own subject.

The rule to hold all lesson: a criterion is not accepted unless a question is attached to it. “Magnitude” on its own is a word. “A historian estimating turnout needs magnitude” is the move.